

LOW RANK p-CELLS

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1. WHAT THIS NOTE RECORDS

This is a companion to the low rank note on ordinary Kazhdan–Lusztig cells. I use 0-cell for the usual Kazhdan–Lusztig two-sided cells, and p-cell for the two-sided cells obtained from the p -canonical basis in the computations.

The point of this file is to be a data dump with some formatting: the number of cells, their sizes, their compressed matrices, and the guessed diagonal asymptotic category pieces. The guesses are included because this file is meant for the GitHub record, not for the conservative appendix in the paper.

Matrix convention. The matrices below are displayed in compressed square block form. If

$$r = (r_1, \dots, r_a), \quad M = (m_{ij}) \in M_{a \times a}(\mathbb{Z}_{\geq 0}),$$

then $[r; M]$ denotes the square matrix obtained from M by replacing m_{ij} by a constant $r_i \times r_j$ block with all entries equal to m_{ij} . For example,

$$\left[(2, 3); \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \right] = \begin{pmatrix} 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

Rows and columns have been permuted to group equal profiles. Strongly regular cells have matrix $1_{n,n}$, so they are usually not listed in the detailed matrix catalogue.

Matrix entries are color-coded: 0, 1, 2, 3, 4, 5, 6, 7, 9. The colors are only visual aids; they do not add extra structure.

Asymptotic category convention.

The ordinary low rank note labels diagonal asymptotic categories such as Vect , $\text{Rep}(C_2)$, $\text{Rep}(C_3)$, $\text{Rep}(S_3)$, Vec_{C_3} , or similar pointed versions. In characteristic p , the relevant bookkeeping is no longer semisimple. I therefore write the expected diagonal p -asymptotic piece next to each non-strongly-regular matrix.

The basic reduction rule used below is

$$\text{Rep}_{\mathbb{C}}(C_p) \rightsquigarrow \text{Vect} \text{ and } \text{Proj}(kC_p),$$

where k has characteristic p . Thus an ordinary C_2 -piece at $p = 2$ splits into a trivial piece Vect and a projective piece $\text{Proj}(kC_2)$. Diagonal size 4 at $p = 2$ is recorded as $\text{Proj}(k(C_2)^2)$. Away from the defining prime I keep the ordinary semisimple label, for instance $\text{Rep}(C_2)$ at $p = 3$.

For larger exceptional cells I use warning labels when the matrix suggests a diagonal piece but the categorical splitting has not been independently identified. These warnings are intentional guesses, not theorem statements.

A change of p -cells is stronger than the existence of torsion in an intersection form. Torsion is expected, and known, for many primes in sufficiently high rank. This file only records where the p -cells themselves change in the available low rank data, together with the characteristic zero reference rows and the raw-file sanity checks.

2. INVENTORY

Here is a summary table. The characteristic zero rows are included for comparison. Positive characteristic rows are listed when they occur in the raw files, with the main changed rows shown in the body and the remaining sanity checks also recorded in the file inventory.

type	char.	# cells	non-sr cells	max entry
$B_2 = C_2$	0	3	1	2
B_2	2	4	1	2
C_2	2	4	1	2
$B_3 = C_3$	0	6	2	2
B_3	2	8	2	2
C_3	2	8	2	2
$B_4 = C_4$	0	10	5	2
B_4	2	15	5	2
C_4	2	15	5	2
$B_5 = C_5$	0	16	10	2
B_5	2	26	10	2
C_5	2	26	10	2
$B_6 = C_6$	0	26	15	4
B_6	2	45	18	4
C_6	2	39	17	4
D_4	0	11	1	2
D_4	2	12	1	2
D_5	0	14	2	2
D_5	2	16	2	2
D_6	0	27	5	2
D_6	2	32	5	2
G_2	0	3	1	3
G_2	2	4	1	2
G_2	3	4	1	3
F_4	0	11	3	9
F_4	2	17	5	7
F_4	3	12	3	9
F_4	5	11	3	9
E_6	0	17	3	6
E_6	2	20	3	6
E_6	3	18	3	6
E_6	5	17	3	6

Here non-sr means non-strongly-regular. Types B and C are listed separately in positive characteristic. Although they have the same Coxeter group, the p -canonical basis depends on the realization, and the two root lengths can matter in small characteristic. This is visible in rank six: the computed 2-cell decompositions of types B_6 and C_6 have different numbers of two-sided 2-cells.

3. CELL SIZE VECTORS

The following tables give the cell size vectors in the order produced by the computations. This is not the cell order in general. Characteristic zero rows are included for comparison.

3.1. Classical types.

type	char.	cell sizes
$B_2 = C_2$	0	1, 6, 1
B_2	2	1, 5, 1, 1
C_2	2	1, 5, 1, 1
$B_3 = C_3$	0	1, 14, 9, 9, 14, 1
B_3	2	1, 10, 9, 4, 9, 13, 1, 1
C_3	2	1, 13, 9, 9, 10, 1, 4, 1
$B_4 = C_4$	0	1, 26, 56, 64, 54, 36, 64, 56, 26, 1
B_4	2	1, 17, 9, 52, 64, 36, 45, 64, 4, 40, 9, 16, 25, 1, 1
C_4	2	1, 25, 40, 16, 64, 36, 45, 64, 52, 1, 9, 17, 4, 9, 1
$B_5 = C_5$	0	1, 42, 150, 100, 225, 152, 600, 650, 650, 600, 152, 225, 100, 150, 42, 1
B_5	2	1, 26, 16, 125, 225, 100, 500, 25, 100, 116, 625, 425, 36, 225, 25, 500, 100, 100, 136, 225, 125, 16, 25, 41, 1, 1
C_5	2	1, 41, 125, 225, 100, 500, 136, 425, 625, 25, 100, 225, 500, 100, 100, 1, 16, 25, 116, 225, 125, 36, 25, 26, 16, 1
$B_6 = C_6$	0	1, 62, 342, 576, 650, 3150, 350, 1600, 2432, 3402, 900, 2025, 14500, 600, 2025, 900, 3402, 2432, 1600, 350, 576, 3150, 650, 342, 62, 1
B_6	2	1, 37, 25, 261, 576, 625, 2250, 81, 900, 1600, 900, 250, 2176, 2025, 2106, 10225, 25, 925, 100, 1296, 2425, 256, 900, 900, 2025, 1600, 500, 3321, 1856, 2925, 25, 425, 100, 576, 225, 325, 576, 81, 225, 306, 25, 36, 61, 1, 1
C_6	2	1, 62, 992, 576, 4750, 900, 325, 1856, 3321, 2025, 10225, 576, 2425, 925, 900, 900, 2025, 1600, 500, 2106, 2176, 1296, 2250, 900, 625, 25, 81, 25, 100, 256, 25, 250, 576, 100, 261, 81, 37, 25, 1
D_4	0	1, 16, 9, 9, 9, 104, 9, 9, 9, 16, 1
D_4	2	1, 16, 9, 9, 9, 100, 4, 9, 9, 9, 16, 1
D_5	0	1, 25, 16, 100, 350, 400, 100, 36, 400, 350, 100, 16, 25, 1
D_5	2	1, 25, 16, 100, 325, 25, 400, 100, 36, 400, 325, 25, 100, 16, 25, 1
D_6	0	1, 36, 225, 25, 882, 100, 100, 3650, 900, 2025, 900, 100, 1600, 1600, 1952, 900, 100, 900, 2025, 3650, 882, 100, 100, 225, 25, 36, 1
D_6	2	1, 36, 225, 25, 882, 100, 801, 81, 3625, 900, 2025, 900, 100, 1696, 256, 1952, 900, 100, 900, 2025, 3625, 882, 801, 81, 100, 100, 225, 25, 36, 1

3.2. Exceptional types.

type	char.	cell sizes
G_2	0	1, 10, 1
G_2	2	1, 6, 4, 1
G_2	3	1, 9, 1, 1
F_4	0	1, 24, 81, 64, 64, 684, 64, 64, 81, 24, 1
F_4	2	1, 20, 81, 64, 64, 298, 272, 4, 97, 16, 64, 64, 81, 1, 20, 4, 1
F_4	3	1, 24, 81, 64, 64, 648, 36, 64, 64, 81, 24, 1
F_4	5	1, 24, 81, 64, 64, 684, 64, 64, 81, 24, 1
E_6	0	1, 36, 400, 1350, 4096, 3600, 6561, 576, 18600, 576, 6561, 3600, 4096, 1350, 400, 36, 1
E_6	2	1, 36, 400, 1125, 225, 4096, 3600, 6561, 576, 15000, 3600, 6561, 3600, 576, 4096, 1125, 225, 400, 36, 1
E_6	3	1, 36, 400, 1350, 4096, 3600, 6561, 576, 18500, 100, 6561, 3600, 576, 4096, 1350, 400, 36, 1
E_6	5	1, 36, 400, 1350, 4096, 3600, 6561, 576, 18600, 6561, 3600, 576, 4096, 1350, 400, 36, 1

4. COMPRESSED NON-STRONGLY-REGULAR MATRICES

The classical characteristic zero matrices are mostly of the form

$$U(a, b) = \left[(a, b); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right],$$

while most changed classical 2-cells have the compressed matrix shape

$$T(a, b) = \left[(a, b); \mid \begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right].$$

The number before the colon is the size of the corresponding cell.

4.1. Summary table for the $U(a, b)$ and $T(a, b)$ families.

type	char.	non-strongly-regular cells of type $U(a, b)$ or $T(a, b)$
$B_2 = C_2$	0	6 : $U(1, 1)$
B_2	2	5 : $T(1, 1)$
C_2	2	5 : $T(1, 1)$
$B_3 = C_3$	0	14 : $U(1, 2)$, 14 : $U(2, 1)$
B_3	2	10 : $T(2, 1)$, 13 : $T(1, 2)$
C_3	2	13 : $T(1, 2)$, 10 : $T(2, 1)$
$B_4 = C_4$	0	26 : $U(1, 3)$, 56 : $U(2, 4)$, 54 : $U(3, 3)$, 56 : $U(4, 2)$, 26 : $U(3, 1)$
B_4	2	17 : $T(3, 1)$, 52 : $T(2, 4)$, 45 : $T(3, 3)$, 40 : $T(4, 2)$, 25 : $T(1, 3)$
C_4	2	25 : $T(1, 3)$, 40 : $T(4, 2)$, 45 : $T(3, 3)$, 52 : $T(2, 4)$, 17 : $T(3, 1)$
$B_5 = C_5$	0	42 : $U(1, 4)$, 150 : $U(5, 5)$, 152 : $U(4, 6)$, 600 : $U(10, 10)$, 650 : $U(5, 15)$, 650 : $U(15, 5)$, 600 : $U(10, 10)$ 152 : $U(6, 4)$, 150 : $U(5, 5)$, 42 : $U(4, 1)$
B_5	2	26 : $T(4, 1)$, 125 : $T(5, 5)$, 500 : $T(10, 10)$, 116 : $T(6, 4)$, 625 : $T(5, 15)$, 425 : $T(15, 5)$, 500 : $T(10, 10)$ 136 : $T(4, 6)$, 125 : $T(5, 5)$, 41 : $T(1, 4)$
C_5	2	41 : $T(1, 4)$, 125 : $T(5, 5)$, 500 : $T(10, 10)$, 136 : $T(4, 6)$, 425 : $T(15, 5)$, 625 : $T(5, 15)$, 500 : $T(10, 10)$ 116 : $T(6, 4)$, 125 : $T(5, 5)$, 26 : $T(4, 1)$
$B_6 = C_6$	0	62 : $U(1, 5)$, 342 : $U(6, 9)$, 650 : $U(5, 15)$, 3150 : $U(15, 30)$, 350 : $U(5, 10)$, 2432 : $U(16, 24)$ 3402 : $U(9, 36)$, 600 : $U(10, 10)$, 2432 : $U(24, 16)$, 350 : $U(10, 5)$, 3150 : $U(30, 15)$, 650 : $U(15, 5)$ 342 : $U(9, 6)$, 62 : $U(5, 1)$
B_6	2	37 : $T(5, 1)$, 261 : $T(9, 6)$, 625 : $T(5, 15)$, 2250 : $T(30, 15)$, 250 : $T(10, 5)$, 2176 : $T(16, 24)$ 2106 : $T(36, 9)$, 925 : $T(25, 5)$, 2425 : $T(25, 20)$, 500 : $T(10, 10)$, 3321 : $T(9, 36)$, 1856 : $T(24, 16)$ 2925 : $T(15, 30)$, 425 : $T(15, 5)$, 325 : $T(5, 10)$, 306 : $T(6, 9)$, 61 : $T(1, 5)$
C_6	2	325 : $T(5, 10)$, 1856 : $T(24, 16)$, 3321 : $T(9, 36)$, 2425 : $T(25, 20)$, 925 : $T(25, 5)$, 500 : $T(10, 10)$ 2106 : $T(36, 9)$, 2176 : $T(16, 24)$, 2250 : $T(30, 15)$, 625 : $T(5, 15)$, 250 : $T(10, 5)$, 261 : $T(9, 6)$ 37 : $T(5, 1)$
D_4	0	104 : $U(2, 6)$
D_4	2	100 : $T(2, 6)$
D_5	0	350 : $U(5, 10)$, 350 : $U(5, 10)$
D_5	2	325 : $T(5, 10)$, 325 : $T(5, 10)$
D_6	0	882 : $U(9, 15)$, 3650 : $U(5, 40)$, 1952 : $U(16, 20)$, 3650 : $U(5, 40)$, 882 : $U(9, 15)$
D_6	2	801 : $T(9, 15)$, 3625 : $T(5, 40)$, 1696 : $T(16, 20)$, 3625 : $T(5, 40)$, 801 : $T(9, 15)$

4.2. Expanded catalogue for the U and T families.

$B_2 = C_2$ in characteristic 0.

size 6: $U(1,1)$

$\mathcal{A}_\Delta$: $\text{Rep}(C_2)$ -type.

$$\left[(1, 1); \mid \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right]$$

B_2 in characteristic 2.

size 5: $T(1,1)$

$\mathcal{A}_\Delta$: Vect , $\text{Proj}(kC_2)$.

$$\left[(1, 1); \mid \begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right]$$

C_2 in characteristic 2.

size 5: T(1,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1,1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

$B_3 = C_3$ in characteristic 0.

size 14: U(1,2)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(1,2); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 14: U(2,1)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(2,1); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

B_3 in characteristic 2.

size 10: T(2,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(2,1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 13: T(1,2)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1,2); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

C_3 in characteristic 2.

size 13: T(1,2)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1,2); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 10: T(2,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(2,1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

$B_4 = C_4$ in characteristic 0.

size 26: U(1,3)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(1, 3); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 56: U(2,4)

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ -type.

$$\left[(2, 4); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 54: U(3,3)

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ -type.

$$\left[(3, 3); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 56: U(4,2)

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ -type.

$$\left[(4, 2); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 26: U(3,1)

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ -type.

$$\left[(3, 1); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

B_4 in characteristic 2.

size 17: T(3,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2)$.

$$\left[(3, 1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 52: T(2,4)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2)$.

$$\left[(2, 4); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 45: T(3,3)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2)$.

$$\left[(3, 3); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 40: T(4,2)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2)$.

$$\left[(4, 2); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 25: T(1,3) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1,3); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

 C_4 in characteristic 2.**size 25: T(1,3)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1,3); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 40: T(4,2) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(4,2); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 45: T(3,3) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(3,3); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 52: T(2,4) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(2,4); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 17: T(3,1) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(3,1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

 $B_5 = C_5$ in characteristic 0.**size 42: U(1,4)** $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(1,4); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 150: U(5,5) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5,5); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 152: U(4,6)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(4, 6); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 600: U(10,10)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(10, 10); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 650: U(5,15)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(5, 15); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 650: U(15,5)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(15, 5); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 600: U(10,10)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(10, 10); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 152: U(6,4)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(6, 4); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 150: U(5,5)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(5, 5); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

size 42: U(4,1)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(4, 1); \left| \begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right| \right]$$

B_5 in characteristic 2.

size 26: T(4,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(4, 1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 125: T(5,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 5); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 500: T(10,10)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 116: T(6,4)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(6, 4); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 625: T(5,15)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 15); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 425: T(15,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(15, 5); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 500: T(10,10)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 136: T(4,6)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(4, 6); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 125: T(5,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 5); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 41: T(1,4) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1, 4); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

 C_5 in characteristic 2.**size 41: T(1,4)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1, 4); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 125: T(5,5) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 5); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 500: T(10,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 136: T(4,6) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(4, 6); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 425: T(15,5) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(15, 5); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 625: T(5,15) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 15); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 500: T(10,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 116: T(6,4) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(6, 4); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 125: T(5,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 5); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 26: T(4,1)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(4, 1); \left(\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array} \right) \right]$$

$B_6 = C_6$ in characteristic 0.

size 62: U(1,5)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(1, 5); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 342: U(6,9)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(6, 9); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 650: U(5,15)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 15); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 3150: U(15,30)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(15, 30); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 350: U(5,10)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 10); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 2432: U(16,24)

$\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(16, 24); \left(\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right) \right]$$

size 3402: U(9,36) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(9, 36); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 600: U(10,10) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(10, 10); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 2432: U(24,16) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(24, 16); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 350: U(10,5) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(10, 5); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 3150: U(30,15) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(30, 15); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 650: U(15,5) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(15, 5); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 342: U(9,6) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(9, 6); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 62: U(5,1) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type.}$

$$\left[(5, 1); \begin{array}{|c|c|c|} \hline 2 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

*B₆ in characteristic 2.***size 37: T(5,1)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 1); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 261: T(9,6)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 6); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 625: T(5,15)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 15); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2250: T(30,15)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(30, 15); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 250: T(10,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 5); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2176: T(16,24)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(16, 24); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2106: T(36,9)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(36, 9); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 925: T(25,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(25, 5); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2425: T(25,20)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(25, 20); \begin{array}{|cc|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 500: T(10,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 3321: T(9,36) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 36); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 1856: T(24,16) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(24, 16); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 2925: T(15,30) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(15, 30); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 425: T(15,5) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(15, 5); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 325: T(5,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 10); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 306: T(6,9) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(6, 9); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

size 61: T(1,5) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(1, 5); \begin{array}{|c|c|c|} \hline 1 & 1 & \\ \hline 1 & 2 & \\ \hline \end{array} \right]$$

 C_6 in characteristic 2.**size 325: T(5,10)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 10); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 1856: T(24,16)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(24, 16); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 3321: T(9,36)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 36); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2425: T(25,20)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(25, 20); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 925: T(25,5)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(25, 5); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 500: T(10,10)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 10); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2106: T(36,9)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(36, 9); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2176: T(16,24)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(16, 24); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 2250: T(30,15)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(30, 15); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 625: T(5,15) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 15); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 250: T(10,5) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(10, 5); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 261: T(9,6) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 6); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 37: T(5,1) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 1); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

*D₄ in characteristic 0.***size 104: U(2,6)** $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(2, 6); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

*D₄ in characteristic 2.***size 100: T(2,6)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(2, 6); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

*D₅ in characteristic 0.***size 350: U(5,10)** $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 10); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 350: U(5,10) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 10); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

D₅ in characteristic 2.

size 325: T(5,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 10); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 325: T(5,10) $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 10); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

*D₆ in characteristic 0.***size 882: U(9,15)** $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(9, 15); \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 3650: U(5,40) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 40); \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 1952: U(16,20) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(16, 20); \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 3650: U(5,40) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(5, 40); \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 882: U(9,15) $\mathcal{A}_\Delta : \text{Rep}(C_2)\text{-type}.$

$$\left[(9, 15); \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

*D₆ in characteristic 2.***size 801: T(9,15)** $\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 15); \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \right]$$

size 3625: T(5,40)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 40); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 1696: T(16,20)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(16, 20); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 3625: T(5,40)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(5, 40); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

size 801: T(9,15)

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2).$

$$\left[(9, 15); \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

4.3. Additional classical rank-six matrices. In characteristic zero, the $B_6 = C_6$ cell of size 14500 has compressed matrix

characteristic zero, size 14500

$\mathcal{A}_\Delta : \text{Rep}((C_2)^2)\text{-type} / \text{pointed } (C_2)^2\text{-type}.$

$$\left[(5, 5, 20, 25, 25); \begin{pmatrix} 4 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & 4 & 2 & 2 \\ 2 & 2 & 2 & 4 & 1 \\ 2 & 2 & 2 & 1 & 4 \end{pmatrix} \right]$$

In type B_6 , the 2-cell of size 10225 has compressed matrix

$B_6, p = 2$, size 10225

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2), \text{Proj}(k(C_2)^2).$

$$\left[(25, 5, 5, 20, 25); \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 \end{pmatrix} \right]$$

In type C_6 , the corresponding 2-cell of size 10225 has compressed matrix

$C_6, p = 2$, size 10225

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2), \text{Proj}(k(C_2)^2).$

$$\left[(25, 5, 5, 20, 25); \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 \end{pmatrix} \right]$$

The remaining non- $T(a, b)$ matrices in type C_6 are

$C_6, p = 2$, size **62**

$\mathcal{A}_\Delta : \text{Proj}(kC_2)$ -type (**warning**).

$$\left[(1, 5); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

$C_6, p = 2$, size **992**

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2), \text{Proj}(k(C_2)^2)$ -type (**warning**).

$$\left[(2, 2, 9, 5, 7, 6); \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 2 & 4 \end{pmatrix} \right]$$

$C_6, p = 2$, size **4750**

$\mathcal{A}_\Delta : \text{Vect}, \text{Proj}(kC_2), \text{unknown}_3$ (**warning**).

$$\left[(12, 37, 12, 30, 3); \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 3 \end{pmatrix} \right]$$

4.4. Exceptional matrices with guessed category labels. The exceptional cells are where the labels are most speculative. The point of this GitHub file is to record these guesses rather than hide them.

$G_2, p = 0$, size **10**

$\mathcal{A}_\Delta : SO(3)_6$ -type.

$$\left[(1, 1); \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \right]$$

$G_2, p = 2$, size **6**

$\mathcal{A}_\Delta : \text{Proj}(kC_2)$.

$$\left[(1, 1); \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \right]$$

$G_2, p = 3$, size **9**

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ (semisimple at $p = 3$), $\text{Proj}(kC_3)$.

$$\left[(1, 1); \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix} \right]$$

$F_4, p = 0$, size **684**

$\mathcal{A}_\Delta : \text{Rep}(S_4)$ -type, G -type, G' -type.

$$\left[(3, 3, 4, 1, 1); \begin{pmatrix} 5 & 3 & 4 & 5 & 2 \\ 3 & 5 & 4 & 2 & 5 \\ 4 & 4 & 9 & 6 & 6 \\ 5 & 2 & 6 & 9 & 3 \\ 2 & 5 & 6 & 3 & 9 \end{pmatrix} \right]$$

$F_4, p = 2$, size **298**

$\mathcal{A}_\Delta : \text{Proj}(kC_2), \text{Vect}, \text{unknown}_3, \text{unknown}_7$ (warning).

$$\left[(3, 1, 3, 4, 1); \begin{pmatrix} 2 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 1 & 2 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 & 7 \end{pmatrix} \right]$$

$F_4, p = 3$, size **648**

$\mathcal{A}_\Delta : \text{unknown}_4$ (warning), $\text{Proj}(k(C_3)^2)$.

$$\left[(3, 3, 1, 1, 4); \begin{pmatrix} 4 & 2 & 2 & 5 & 4 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 9 & 3 & 6 \\ 5 & 2 & 3 & 9 & 6 \\ 4 & 4 & 6 & 6 & 9 \end{pmatrix} \right]$$

$E_6, p = 0$, size **18600**

$\mathcal{A}_\Delta : \text{Rep}(S_3), \text{Rep}(S_3), \text{Vec}_{S_3}$.

$$\left[(10, 50, 20); \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 3 & 6 \end{pmatrix} \right]$$

$E_6, p = 2$, size **15000**

$\mathcal{A}_\Delta : \text{Proj}(kC_2), \text{unknown}_6$ (warning).

$$\left[(10, 50, 20); \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix} \right]$$

$E_6, p = 3$, size **18500**

$\mathcal{A}_\Delta : \text{Rep}(C_2)$ (semisimple at $p = 3$), $\text{Proj}(kC_3), \text{unknown}_6$ (warning).

$$\left[(10, 50, 20); \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 3 & 6 \end{pmatrix} \right]$$

5. RAW FILE INVENTORY

The following raw files were used for this summary. The cell sizes are read from the headers of the corresponding JCe11 blocks. The classical odd-prime files are included as sanity checks: their cell-size multisets agree with characteristic zero, although the computed order may differ.

file	total	# cells	non-sr	max entry
b-2-2.txt	8	4	1	2
b-3-2.txt	48	8	2	2
b-4-2.txt	384	15	5	2
b-5-2.txt	3840	26	10	2
b-6-2.txt	46080	45	18	4
b-6-3.txt	46080	26	16	4
c-2-2.txt	8	4	1	2
c-3-2.txt	48	8	2	2
c-4-2.txt	384	15	5	2
c-5-2.txt	3840	26	10	2
c-6-2.txt	46080	39	17	4
c-6-3.txt	46080	26	16	4
d-4-2.txt	192	12	1	2
d-5-2.txt	1920	16	2	2
d-6-2.txt	23040	32	5	2
d-6-3.txt	23040	27	5	2
e-6-2.txt	51840	20	3	6
e-6-3.txt	51840	18	3	6
e-6-5.txt	51840	17	3	6
f-4-2.txt	1152	17	5	7
f-4-3.txt	1152	12	3	9
f-4-5.txt	1152	11	3	9
g-2-2.txt	12	4	1	2
g-2-3.txt	12	4	1	3

More data, including raw files and multiplication tables, is intended to live with [1]. The table above is just an index of the files summarized here.

REFERENCES

- [1] V. Miemietz, M. Roth, and D. Tubbenhauer. Low-rank p -cell data and accompanying code. GitHub repository.